

VK-RCD：面向隐私友好与任意模态缺失的人体感知蒸馏框架

基于官方 X-Fi 的系统对比

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2026-05-09

摘要

多模态人体感知需要在高精度、隐私保护和真实部署鲁棒性之间取得平衡。官方 X-Fi 证明了单一模型处理多种传感器组合的可行性，但其视觉分支仍依赖原始 RGB 图像，且缺失模态下的可靠决策仍有提升空间。本文提出 VK-RCD，以视觉关键点（VK）替代原始 RGB，利用全模态 Teacher 学习跨模态人体结构知识，并通过随机缺失模态训练将知识蒸馏到 Student-VK 与非视觉 Student-NV。与官方 X-Fi 相比，本文 Student-VK 在 MM-Fi HPE 全模态设置下将 MPJPE 从 83.70 mm 降至 50.17 mm（+40.06%），在 HAR 全模态设置下将准确率从 72.20% 提升至 96.54%（+33.71%）。本文进一步报告 HPE 31 种组合、HAR 15 种组合、非视觉部署、跨场景/跨主体和关键消融，所有已同步指标均纳入正文或附录。

1 引言

人体感知广泛服务于康复评估、智能家居、人机交互和机器人协作。RGB 图像虽然信息丰富，却包含面部、服饰、身份和家庭环境等隐私敏感线索。Depth、LiDAR、mmWave 与 WiFi-CSI 等非 RGB 传感器能够缓解部分隐私风险，但单一模态又容易受到遮挡、点云稀疏、信道变化或环境迁移影响。官方 X-Fi 提出了模态不变人体感知框架，一次训练即可支持参与训练的传感器单独或任意组合使用，并在 MM-Fi 的 HPE/HAR 任务上取得优势。

本文的目标不是简单把 RGB backbone 换成小 MLP，而是重新组织隐私友好人体感知的训练与部署链路：使用 VK 作为结构化视觉替代，保留人体姿态与动作信息；利用全模态 Teacher 作为特权知识源；通过随机缺失模态训练让 Student-VK 在任意可用模态集合下工作；进一步训练 Student-NV 验证无视觉输入的部署能力；最后通过可靠性感知融合削弱低质量模态对最终决策的负面影响。

本文贡献如下：1）提出面向 MM-Fi HPE/HAR 的非 RGB 隐私友好 VK-RCD 框架；2）构建 full-to-missing 的 Teacher-Student 蒸馏流程；3）系统评估任意模态组合、非视觉组合、跨场景/跨主体和 VK 噪声鲁棒性；4）与官方 X-Fi 逐项对齐，并用蓝红标注报告性能提升或退化。

2 相关工作

多模态人体感知。 多模态人体感知融合视觉、深度、点云、毫米波与无线信道信息。不同模态在空间分辨率、弱光鲁棒性、穿透性、成本和隐私保护方面互补。MM-Fi 提供同步多模态数据和 HPE/HAR 任务，X-Fi 则进一步验证了任意模态组合人体感知的可行性。

缺失模态学习。 真实系统中传感器可能因遮挡、权限、硬件故障或通信异常而缺失。固定融合模型通常需要为不同组合重新训练，而模态不变学习、动态融合与蒸馏方法试图让模型在可变输入集合下保持稳定。本文沿用任意组合评估协议，但强调训练阶段随机缺失和 Teacher 到 Student 的知识迁移。

隐私友好与蒸馏。 非 RGB 传感器和骨架化视觉表示都可降低原始图像暴露风险。本文采用 VK 作为结构化视觉输入，同时使用全模态 Teacher 提供软目标和中间结构知识，使最终部署的 Student 不必依赖原始 RGB 图像。

3 方法

3.1 问题定义

给定模态集合 $\mathcal{M} = \{VK, D, L, R, W\}$, 其中 D, L, R, W 分别表示 Depth、LiDAR、mmWave 与 WiFi-CSI。HPE 输出 17 个三维人体关节, HAR 输出 27 类动作概率。对任意可用子集 $\mathcal{S} \subseteq \mathcal{M}$, 模型均需完成预测。表中 I/VK 表示官方 X-Fi 的 I 为 RGB, 而本文对应位置为 VK, 二者不被视作完全同源输入。

3.2 VK 编码与跨模态融合

VK 分支接收 17 个二维关节坐标, 先用共享线性层将每个关节独立升维, 再在关节维和模态维进行全局交互。Depth、LiDAR、mmWave 与 WiFi-CSI 使用各自传感器 backbone 抽取特征后投影到统一 token 空间。融合模块根据当前可用模态集生成跨模态表示, 并由 HPE/HAR 任务头输出预测。

3.3 Teacher-Student 蒸馏与随机缺失

Teacher 使用全模态训练, 学习完整传感器条件下的人体结构表示。Student-VK 在训练时随机采样可用模态子集, 并同时优化监督损失、输出蒸馏损失和特征蒸馏损失。Student-NV 移除 VK, 仅使用非视觉传感器, 以检验完全不依赖视觉输入时的部署性能。可靠性感知融合为不同模态分配动态权重; Uniform 消融使用等权融合, NoKD 消融移除 Teacher 蒸馏。

4 实验设置

实验基于 MM-Fi。HPE 使用 VK、Depth、LiDAR、mmWave、WiFi-CSI 五类输入, 指标为 MPJPE 与 PA-MPJPE, 单位为 mm, 越低越好。HAR 使用 VK、Depth、LiDAR、mmWave, 指标为 Accuracy 与 Macro-F1, 越高越好。训练遵循官方截断风格, 评估使用完整验证集。所有官方数值来自 X-Fi 论文公开表格或本地整理的 legacy CSV。

颜色规则如下: HPE 的 $\Delta = \text{Ours} - \text{Official}$, 负数为更好; HAR 的 $\Delta = \text{Ours} - \text{Official}$, 正数为更好。蓝色表示本文结果优于官方, 红色表示退化。

5 定量结果

5.1 HPE 全模态与官方对比

表 1: HPE 全模态结果。Student-NV 不含 VK, 因此与官方非视觉 D+L+R+W 组合比较。

方法	组合	Ours MPJPE	X-Fi MPJPE	Δ	Rel.Imp	Ours PA	Δ PA	Rel.Imp
Baseline	I/VK+D+L+R+W	49.14	83.70	-34.56	+41.30%	33.48	-14.12	+29.66%
Teacher	I/VK+D+L+R+W	48.45	83.70	-35.25	+42.11%	33.06	-14.54	+30.54%
Student-VK	I/VK+D+L+R+W	50.17	83.70	-33.53	+40.06%	33.48	-14.12	+29.66%
Student-NV	D+L+R+W	51.03	97.60	-46.57	+47.72%	35.69	-11.71	+24.69%

表 2: HPE Student-VK 按可用模态数量的 31 组合平均结果。

可用模态数	组合数	Ours MPJPE	X-Fi MPJPE	Rel.Imp	Ours PA	X-Fi PA	Rel.Imp
1	5	124.71	143.16	+12.89%	67.99	77.40	+12.15%
2	10	78.09	104.97	+25.61%	46.85	59.72	+21.55%
3	10	60.78	92.58	+34.35%	38.09	52.32	+27.20%
4	5	53.54	87.94	+39.12%	34.89	49.58	+29.62%
5	1	50.17	83.70	+40.06%	33.48	47.60	+29.66%
平均	31	75.17	103.70	+27.52%	45.07	58.16	+22.50%

5.2 HAR 主结果与跨协议泛化

表 3: HAR random split 全模态、非视觉与消融结果。

方法	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
Teacher	I/VK+D+L+R	95.57	72.20	+23.37	+32.37%	95.56
Student-VK	I/VK+D+L+R	96.54	72.20	+24.34	+33.71%	96.53
Student-NV	D+L+R	96.12	80.50	+15.62	+19.41%	96.13
NoKD	I/VK+D+L+R	95.97	72.20	+23.77	+32.92%	95.95
Uniform	I/VK+D+L+R	95.40	72.20	+23.20	+32.14%	95.40

表 4: HAR random/cross-scene/cross-subject 全模态与非视觉结果。

协议	方法	组合	Acc	Macro-F1	X-Fi Acc	Δ	Rel.Imp
random	Teacher	I/VK+D+L+R	95.57	95.56	72.20	+23.37	+32.37%
random	Student-VK	I/VK+D+L+R	96.54	96.53	72.20	+24.34	+33.71%
random	Student-NV	D+L+R	96.12	96.13	80.50	+15.62	+19.41%
cross-scene	Teacher	I/VK+D+L+R	94.80	94.83	72.20	+22.60	+31.31%
cross-scene	Student-VK	I/VK+D+L+R	96.12	96.10	72.20	+23.92	+33.14%
cross-scene	Student-NV	D+L+R	95.41	95.40	80.50	+14.91	+18.52%
cross-subject	Teacher	I/VK+D+L+R	96.12	96.10	72.20	+23.92	+33.14%
cross-subject	Student-VK	I/VK+D+L+R	96.12	96.10	72.20	+23.92	+33.14%
cross-subject	Student-NV	D+L+R	95.66	95.64	80.50	+15.16	+18.83%

6 消融与鲁棒性

HAR 消融显示, Student-VK 在全模态 random split 上达到 96.54% Acc, 高于 NoKD 与 Uniform 的对应结果, 说明蒸馏和可靠性感知融合都对最终部署模型有贡献。Student-NV 在无 VK 的情况下仍保持强性能, 说明非视觉传感器组合能够承担隐私友好部署场景。

表 5: HPE VK 噪声与关节缺失鲁棒性。

噪声标准差	关节丢失率	MPJPE	PA-MPJPE	MSE
0.0	0.0	50.18	33.48	0.001459
0.0	0.1	52.64	34.57	0.001571
0.0	0.3	54.10	35.19	0.001638
0.0	0.5	54.75	35.50	0.001669
5.0	0.0	50.88	33.92	0.001488
5.0	0.1	52.88	34.71	0.001582
5.0	0.3	54.12	35.19	0.001640
5.0	0.5	54.76	35.50	0.001670
10.0	0.0	52.08	34.57	0.001539
10.0	0.1	53.26	34.93	0.001599
10.0	0.3	54.12	35.20	0.001638
10.0	0.5	54.74	35.52	0.001669
20.0	0.0	53.63	35.20	0.001611
20.0	0.1	53.85	35.15	0.001621
20.0	0.3	54.21	35.25	0.001644
20.0	0.5	54.78	35.54	0.001670

7 讨论

第一，VK-RCD 的核心创新不是“RGB 换成 VK MLP”，而是将隐私友好输入、缺失模态训练、全模态知识蒸馏和动态可靠性融合组合成完整部署范式。第二，Teacher 在附录的缺失模态评估中可能明显退化，这是因为 Teacher 不是缺失模态部署模型；其作用是为 Student 提供全模态知识。第三，当前本地 HPE cross-scene/cross-subject 与部分 HPE 消融仍未同步，因此本文仅在审计表中列出，不将其写成已完成结论。

8 结论

本文基于 X-Fi 和 MM-Fi 构建 VK-RCD，在不使用原始 RGB 的前提下实现 HPE/HAR 任意模态组合人体感知。当前结果显示，Student-VK 在 HPE 31 组合和 HAR 15 组合上均表现稳定，并在全模态与多数缺失组合上超过官方 X-Fi。后续工作应补齐 HPE 跨场景/跨主体与更完整可靠性消融，以进一步支撑高等级投稿。

A HPE Student-VK 31 组合官方对比

组合	Ours MPJPE	X-Fi MPJPE	Δ	Rel.Imp	Ours PA	X-Fi PA	Δ PA	Rel.Imp
D	53.88	101.80	-47.92	+47.07%	35.61	48.40	-12.79	+26.43%
L	139.90	167.10	-27.20	+16.28%	93.44	103.20	-9.76	+9.46%
R	114.98	127.40	-12.42	+9.75%	58.13	69.80	-11.67	+16.72%
I/VK	92.66	93.90	-1.24	+1.32%	42.60	60.30	-17.70	+29.35%
W	222.14	225.60	-3.46	+1.53%	110.19	105.30	+4.89	-4.64%
D+L	53.00	102.50	-49.50	+48.29%	35.35	48.40	-13.05	+26.96%
D+R	52.32	98.00	-45.68	+46.61%	34.64	47.30	-12.66	+26.77%
D+W	53.54	101.80	-48.26	+47.41%	35.52	48.10	-12.58	+26.15%
L+R	97.54	109.80	-12.26	+11.17%	55.09	63.40	-8.31	+13.11%
L+W	134.11	159.50	-25.39	+15.92%	93.22	102.70	-9.48	+9.23%
R+W	108.54	117.20	-8.66	+7.39%	57.27	62.70	-5.43	+8.66%
I/VK+D	51.65	86.10	-34.45	+40.01%	34.29	48.10	-13.81	+28.71%
I/VK+L	73.18	93.00	-19.82	+21.31%	41.92	59.70	-17.78	+29.78%
I/VK+R	68.93	88.80	-19.87	+22.38%	38.90	57.30	-18.40	+32.11%
I/VK+W	88.11	93.00	-4.89	+5.26%	42.30	59.50	-17.20	+28.91%
D+L+R	51.92	96.00	-44.08	+45.92%	34.50	47.30	-12.80	+27.06%
D+L+W	52.86	102.00	-49.14	+48.18%	35.38	48.10	-12.72	+26.44%
D+R+W	52.13	97.00	-44.87	+46.26%	34.62	47.10	-12.48	+26.50%
L+R+W	94.20	107.40	-13.20	+12.29%	55.10	63.10	-8.00	+12.68%
I/VK+D+L	51.10	84.80	-33.70	+39.74%	34.16	48.20	-14.04	+29.13%
I/VK+D+R	50.50	83.40	-32.90	+39.45%	33.52	47.30	-13.78	+29.13%
I/VK+D+W	51.63	85.30	-33.67	+39.47%	34.29	48.10	-13.81	+28.71%
I/VK+L+R	66.08	88.40	-22.32	+25.25%	38.77	57.20	-18.43	+32.22%
I/VK+L+W	70.68	93.00	-22.32	+24.00%	41.76	59.70	-17.94	+30.05%
I/VK+R+W	66.69	88.50	-21.81	+24.64%	38.80	57.10	-18.30	+32.05%
D+L+R+W	51.78	97.60	-45.82	+46.95%	34.54	47.40	-12.86	+27.13%
I/VK+D+L+R	50.25	83.50	-33.25	+39.82%	33.45	47.60	-14.15	+29.73%
I/VK+D+L+W	51.05	86.00	-34.95	+40.64%	34.18	48.20	-14.02	+29.09%
I/VK+D+R+W	50.41	84.00	-33.59	+39.99%	33.54	47.60	-14.06	+29.54%
I/VK+L+R+W	64.22	88.60	-24.38	+27.52%	38.75	57.10	-18.35	+32.14%

组合	Ours MPJPE	X-Fi MPJPE	Δ	Rel.Imp	Ours PA	X-Fi PA	Δ PA	Rel.Imp
I/VK+D+L+R+W	50.17	83.70	-33.53	+40.06%	33.48	47.60	-14.12	+29.66%

B HPE Teacher 31 组合官方对比

Teacher 不是缺失模态部署模型，以下表格仅用于说明全模态 Teacher 在部分模态输入下存在明显分布偏移。

组合	Ours MPJPE	X-Fi MPJPE	Δ	Rel.Imp	Ours PA	X-Fi PA	Δ PA	Rel.Imp
D	234.48	101.80	+132.68	-130.33%	84.60	48.40	+36.20	-74.79%
L	1615.42	167.10	+1448.32	-866.74%	387.33	103.20	+284.13	-275.32%
R	4196.64	127.40	+4069.24	-3194.07%	298.72	69.80	+228.92	-327.97%
I/VK	487.99	93.90	+394.09	-419.69%	139.35	60.30	+79.05	-131.09%
W	4173.33	225.60	+3947.73	-1749.88%	378.22	105.30	+272.92	-259.18%
D+L	285.80	102.50	+183.30	-178.83%	87.47	48.40	+39.07	-80.72%
D+R	178.87	98.00	+80.87	-82.52%	65.56	47.30	+18.26	-38.60%
D+W	197.61	101.80	+95.81	-94.12%	74.07	48.10	+25.97	-53.99%
L+R	2157.14	109.80	+2047.34	-1864.61%	387.65	63.40	+324.25	-511.44%
L+W	4007.08	159.50	+3847.58	-2412.28%	384.13	102.70	+281.43	-274.03%
R+W	4675.91	117.20	+4558.71	-3889.68%	273.15	62.70	+210.45	-335.65%
I/VK+D	369.12	86.10	+283.02	-328.71%	97.31	48.10	+49.21	-102.31%
I/VK+L	1072.74	93.00	+979.74	-1053.48%	213.07	59.70	+153.37	-256.90%
I/VK+R	814.71	88.80	+725.91	-817.47%	174.05	57.30	+116.75	-203.75%
I/VK+W	2491.83	93.00	+2398.83	-2579.39%	335.20	59.50	+275.70	-463.36%
D+L+R	310.49	96.00	+214.49	-223.43%	75.92	47.30	+28.62	-60.51%
D+L+W	177.40	102.00	+75.40	-73.92%	66.98	48.10	+18.88	-39.25%
D+R+W	121.88	97.00	+24.88	-25.65%	54.83	47.10	+7.73	-16.41%
L+R+W	4360.45	107.40	+4253.05	-3960.01%	322.27	63.10	+259.17	-410.73%
I/VK+D+L	634.60	84.80	+549.80	-648.35%	112.27	48.20	+64.07	-132.93%
I/VK+D+R	465.16	83.40	+381.76	-457.75%	86.45	47.30	+39.15	-82.77%
I/VK+D+W	100.71	85.30	+15.41	-18.07%	51.47	48.10	+3.37	-7.01%
I/VK+L+R	1035.05	88.40	+946.65	-1070.87%	237.95	57.20	+180.75	-316.00%
I/VK+L+W	4736.11	93.00	+4643.11	-4992.59%	271.23	59.70	+211.53	-354.32%
I/VK+R+W	4769.71	88.50	+4681.21	-5289.50%	255.98	57.10	+198.88	-348.30%
D+L+R+W	112.99	97.60	+15.39	-15.77%	50.72	47.40	+3.32	-7.00%
I/VK+D+L+R	593.34	83.50	+509.84	-610.59%	93.24	47.60	+45.64	-95.88%
I/VK+D+L+W	67.89	86.00	-18.11	+21.06%	40.53	48.20	-7.67	+15.91%
I/VK+D+R+W	71.27	84.00	-12.73	+15.15%	38.91	47.60	-8.69	+18.26%
I/VK+L+R+W	4876.08	88.60	+4787.48	-5403.48%	239.18	57.10	+182.08	-318.88%
I/VK+D+L+R+W	48.45	83.70	-35.25	+42.11%	33.06	47.60	-14.54	+30.55%

C HPE VK 鲁棒性完整表

噪声标准差	关节丢失率	MPJPE	PA-MPJPE	MSE
0.0	0.0	50.18	33.48	0.001459
0.0	0.1	52.64	34.57	0.001571
0.0	0.3	54.10	35.19	0.001638
0.0	0.5	54.75	35.50	0.001669
5.0	0.0	50.88	33.92	0.001488
5.0	0.1	52.88	34.71	0.001582
5.0	0.3	54.12	35.19	0.001640
5.0	0.5	54.76	35.50	0.001670
10.0	0.0	52.08	34.57	0.001539
10.0	0.1	53.26	34.93	0.001599
10.0	0.3	54.12	35.20	0.001638
10.0	0.5	54.74	35.52	0.001669
20.0	0.0	53.63	35.20	0.001611
20.0	0.1	53.85	35.15	0.001621
20.0	0.3	54.21	35.25	0.001644
20.0	0.5	54.78	35.54	0.001670

D HAR random split 15 组合官方对比

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
Student-VK	D	88.02	48.10	+39.92	+83.00%	87.86
Student-VK	L	40.80	52.70	-11.90	-22.59%	40.10
Student-VK	R	85.88	85.70	+0.18	+0.21%	86.06
Student-VK	I/VK	65.31	26.50	+38.81	+146.47%	65.70
Student-VK	D+L	88.52	51.60	+36.92	+71.54%	88.39
Student-VK	D+R	96.45	79.80	+16.65	+20.86%	96.44
Student-VK	L+R	89.51	88.70	+0.81	+0.91%	89.57

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
Student-VK	I/VK+D	89.51	45.30	+44.21	+97.59%	89.36
Student-VK	I/VK+L	68.54	35.20	+33.34	+94.73%	69.05
Student-VK	I/VK+R	92.75	73.40	+19.35	+26.36%	92.82
Student-VK	D+L+R	96.44	80.50	+15.94	+19.80%	96.43
Student-VK	I/VK+D+L	89.62	48.70	+40.92	+84.01%	89.49
Student-VK	I/VK+D+R	96.56	70.70	+25.86	+36.57%	96.55
Student-VK	I/VK+L+R	92.73	77.80	+14.93	+19.19%	92.81
Student-VK	I/VK+D+L+R	96.56	72.20	+24.36	+33.74%	96.55

E HAR cross-scene 与 cross-subject 15 组合

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
cross-scene	D	87.41	48.10	+39.31	+81.73%	87.30
cross-scene	L	19.98	52.70	-32.72	-62.09%	17.51
cross-scene	R	80.30	85.70	-5.40	-6.30%	80.17
cross-scene	I/VK	57.99	26.50	+31.49	+118.82%	57.53
cross-scene	D+L	87.65	51.60	+36.05	+69.86%	87.49
cross-scene	D+R	94.94	79.80	+15.14	+18.97%	94.93
cross-scene	L+R	81.15	88.70	-7.55	-8.51%	81.42
cross-scene	I/VK+D	88.84	45.30	+43.54	+96.11%	88.79
cross-scene	I/VK+L	59.39	35.20	+24.19	+68.72%	58.81
cross-scene	I/VK+R	88.76	73.40	+15.36	+20.93%	88.65
cross-scene	D+L+R	94.87	80.50	+14.37	+17.86%	94.87
cross-scene	I/VK+D+L	88.95	48.70	+40.25	+82.64%	88.88
cross-scene	I/VK+D+R	95.32	70.70	+24.62	+34.82%	95.32
cross-scene	I/VK+L+R	88.33	77.80	+10.53	+13.53%	88.34
cross-scene	I/VK+D+L+R	95.25	72.20	+23.05	+31.92%	95.25
cross-subject	D	90.05	48.10	+41.95	+87.21%	89.96
cross-subject	L	38.46	52.70	-14.24	-27.03%	36.69
cross-subject	R	85.21	85.70	-0.49	-0.58%	85.45
cross-subject	I/VK	65.22	26.50	+38.72	+146.13%	67.65
cross-subject	D+L	90.28	51.60	+38.68	+74.97%	90.17
cross-subject	D+R	95.71	79.80	+15.91	+19.93%	95.68
cross-subject	L+R	88.10	88.70	-0.60	-0.67%	88.14
cross-subject	I/VK+D	91.58	45.30	+46.28	+102.17%	91.51
cross-subject	I/VK+L	68.95	35.20	+33.75	+95.88%	70.90
cross-subject	I/VK+R	93.12	73.40	+19.72	+26.87%	93.12
cross-subject	D+L+R	95.74	80.50	+15.24	+18.94%	95.71
cross-subject	I/VK+D+L	91.72	48.70	+43.02	+88.33%	91.64
cross-subject	I/VK+D+R	96.10	70.70	+25.40	+35.92%	96.07
cross-subject	I/VK+L+R	93.06	77.80	+15.26	+19.62%	93.06
cross-subject	I/VK+D+L+R	96.12	72.20	+23.92	+33.13%	96.10

F HAR Student-NV 非视觉组合

协议	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
random	D	88.57	48.10	+40.47	+84.14%	88.47
random	L	26.12	52.70	-26.58	-50.43%	25.01
random	R	83.37	85.70	-2.33	-2.72%	83.57
random	D+L	88.68	51.60	+37.08	+71.85%	88.56
random	D+R	96.18	79.80	+16.38	+20.53%	96.19
random	L+R	85.30	88.70	-3.40	-3.83%	85.38
random	D+L+R	96.12	80.50	+15.62	+19.41%	96.13
cross-scene	D	88.82	48.10	+40.72	+84.65%	88.79
cross-scene	L	8.94	52.70	-43.76	-83.03%	6.39
cross-scene	R	79.70	85.70	-6.00	-7.01%	79.99
cross-scene	D+L	88.92	51.60	+37.32	+72.32%	88.87
cross-scene	D+R	95.45	79.80	+15.65	+19.62%	95.45
cross-scene	L+R	79.68	88.70	-9.02	-10.16%	79.99
cross-scene	D+L+R	95.41	80.50	+14.91	+18.52%	95.40
cross-subject	D	90.30	48.10	+42.20	+87.73%	90.21
cross-subject	L	25.30	52.70	-27.40	-51.99%	22.23
cross-subject	R	84.57	85.70	-1.13	-1.31%	84.75
cross-subject	D+L	90.43	51.60	+38.83	+75.26%	90.34
cross-subject	D+R	95.76	79.80	+15.96	+20.00%	95.74
cross-subject	L+R	86.74	88.70	-1.96	-2.21%	86.76
cross-subject	D+L+R	95.70	80.50	+15.20	+18.88%	95.67

G HAR NoKD 与 Uniform 消融

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
NoKD	D	86.85	48.10	+38.75	+80.55%	86.78
NoKD	L	33.14	52.70	-19.56	-37.11%	31.11
NoKD	R	83.13	85.70	-2.57	-3.00%	83.31

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
NoKD	I/VK	60.94	26.50	+34.44	+129.97%	59.31
NoKD	D+L	87.24	51.60	+35.64	+69.07%	87.25
NoKD	D+R	95.57	79.80	+15.77	+19.76%	95.56
NoKD	L+R	85.92	88.70	-2.78	-3.13%	86.09
NoKD	I/VK+D	88.09	45.30	+42.79	+94.45%	87.94
NoKD	I/VK+L	63.63	35.20	+28.43	+80.75%	62.65
NoKD	I/VK+R	91.05	73.40	+17.65	+24.05%	91.12
NoKD	D+L+R	95.50	80.50	+15.00	+18.64%	95.50
NoKD	I/VK+D+L	88.62	48.70	+39.92	+81.98%	88.53
NoKD	I/VK+D+R	95.96	70.70	+25.26	+35.73%	95.95
NoKD	I/VK+L+R	91.23	77.80	+13.43	+17.27%	91.29
NoKD	I/VK+D+L+R	95.99	72.20	+23.79	+32.96%	95.98
Uniform	D	85.59	48.10	+37.49	+77.93%	85.30
Uniform	L	2.65	52.70	-50.05	-94.97%	1.55
Uniform	R	69.91	85.70	-15.79	-18.42%	68.71
Uniform	I/VK	5.52	26.50	-20.98	-79.16%	1.32
Uniform	D+L	86.16	51.60	+34.56	+66.98%	85.91
Uniform	D+R	95.54	79.80	+15.74	+19.73%	95.54
Uniform	L+R	67.80	88.70	-20.90	-23.56%	66.56
Uniform	I/VK+D	86.48	45.30	+41.18	+90.90%	86.24
Uniform	I/VK+L	4.58	35.20	-30.62	-86.98%	1.76
Uniform	I/VK+R	67.75	73.40	-5.65	-7.69%	66.21
Uniform	D+L+R	95.42	80.50	+14.92	+18.54%	95.42
Uniform	I/VK+D+L	86.43	48.70	+37.73	+77.48%	86.19
Uniform	I/VK+D+R	95.50	70.70	+24.80	+35.07%	95.49
Uniform	I/VK+L+R	67.09	77.80	-10.71	-13.76%	65.08
Uniform	I/VK+D+L+R	95.39	72.20	+23.19	+32.12%	95.39

H HAR 已同步组合结果全集

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
teacher_random	I/VK	3.40	26.50	-23.10	-87.15%	0.45
teacher_random	D	83.71	48.10	+35.61	+74.03%	82.80
teacher_random	L	3.36	52.70	-49.34	-93.63%	1.17
teacher_random	R	66.72	85.70	-18.98	-22.15%	64.78
teacher_random	I/VK+D	83.50	45.30	+38.20	+84.33%	82.60
teacher_random	I/VK+L	3.91	35.20	-31.29	-88.88%	1.31
teacher_random	I/VK+R	66.46	73.40	-6.94	-9.46%	63.67
teacher_random	D+L	83.37	51.60	+31.77	+61.56%	82.54
teacher_random	D+R	95.76	79.80	+15.96	+19.99%	95.74
teacher_random	L+R	66.69	88.70	-22.01	-24.81%	63.96
teacher_random	I/VK+D+L	83.66	48.70	+34.96	+71.79%	82.91
teacher_random	I/VK+D+R	95.66	70.70	+24.96	+35.30%	95.65
teacher_random	I/VK+L+R	66.39	77.80	-11.41	-14.66%	63.36
teacher_random	D+L+R	95.64	80.50	+15.14	+18.80%	95.63
teacher_random	I/VK+D+L+R	95.57	72.20	+23.37	+32.37%	95.56
teacher_cross_scene	I/VK	4.44	26.50	-22.06	-83.25%	0.79
teacher_cross_scene	D	84.86	48.10	+36.76	+76.43%	84.47
teacher_cross_scene	L	3.96	52.70	-48.74	-92.49%	1.50
teacher_cross_scene	R	61.11	85.70	-24.59	-28.69%	59.73
teacher_cross_scene	I/VK+D	85.21	45.30	+39.91	+88.10%	84.97
teacher_cross_scene	I/VK+L	4.01	35.20	-31.19	-88.61%	1.49
teacher_cross_scene	I/VK+R	60.86	73.40	-12.54	-17.09%	59.44
teacher_cross_scene	D+L	84.33	51.60	+32.73	+63.44%	83.84
teacher_cross_scene	D+R	95.07	79.80	+15.27	+19.14%	95.10
teacher_cross_scene	L+R	60.49	88.70	-28.21	-31.81%	58.23
teacher_cross_scene	I/VK+D+L	84.86	48.70	+36.16	+74.26%	84.57
teacher_cross_scene	I/VK+D+R	94.96	70.70	+24.26	+34.32%	94.99
teacher_cross_scene	I/VK+L+R	61.15	77.80	-16.65	-21.40%	59.02
teacher_cross_scene	D+L+R	94.83	80.50	+14.33	+17.81%	94.86
teacher_cross_scene	I/VK+D+L+R	94.80	72.20	+22.60	+31.30%	94.83
teacher_cross_subject	I/VK	5.15	26.50	-21.35	-80.56%	2.00
teacher_cross_subject	D	81.74	48.10	+33.64	+69.94%	80.25
teacher_cross_subject	L	3.89	52.70	-48.81	-92.63%	2.75
teacher_cross_subject	R	72.35	85.70	-13.35	-15.57%	71.76
teacher_cross_subject	I/VK+D	82.87	45.30	+37.57	+82.94%	81.39
teacher_cross_subject	I/VK+L	4.11	35.20	-31.09	-88.33%	1.75
teacher_cross_subject	I/VK+R	69.56	73.40	-3.84	-5.23%	68.95
teacher_cross_subject	D+L	82.45	51.60	+30.85	+59.79%	81.23
teacher_cross_subject	D+R	95.74	79.80	+15.94	+19.98%	95.71
teacher_cross_subject	L+R	69.75	88.70	-18.95	-21.37%	68.99
teacher_cross_subject	I/VK+D+L	82.85	48.70	+34.15	+70.12%	81.62
teacher_cross_subject	I/VK+D+R	95.72	70.70	+25.02	+35.39%	95.69
teacher_cross_subject	I/VK+L+R	69.26	77.80	-8.54	-10.97%	68.44
teacher_cross_subject	D+L+R	95.51	80.50	+15.01	+18.65%	95.47
teacher_cross_subject	I/VK+D+L+R	95.52	72.20	+23.32	+32.30%	95.48
all_combinations.csv	I/VK	3.40	26.50	-23.10	-87.15%	0.45
all_combinations.csv	D	83.71	48.10	+35.61	+74.03%	82.80
all_combinations.csv	L	3.45	52.70	-49.25	-93.46%	1.22
all_combinations.csv	R	66.72	85.70	-18.98	-22.15%	64.78
all_combinations.csv	I/VK+D	83.50	45.30	+38.20	+84.33%	82.60
all_combinations.csv	I/VK+L	3.85	35.20	-31.35	-89.06%	1.26
all_combinations.csv	I/VK+R	66.46	73.40	-6.94	-9.46%	63.67
all_combinations.csv	D+L	83.39	51.60	+31.79	+61.60%	82.57
all_combinations.csv	D+R	95.76	79.80	+15.96	+19.99%	95.74
all_combinations.csv	L+R	66.64	88.70	-22.06	-24.87%	63.92

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
all_combinations.csv	I/VK+D+L	83.70	48.70	+35.00	+71.87%	82.97
all_combinations.csv	I/VK+D+R	95.66	70.70	+24.96	+35.30%	95.65
all_combinations.csv	I/VK+L+R	66.33	77.80	-11.47	-14.74%	63.31
all_combinations.csv	D+L+R	95.62	80.50	+15.12	+18.79%	95.61
all_combinations.csv	I/VK+D+L+R	95.59	72.20	+23.39	+32.39%	95.58
student_vk	I/VK	65.31	26.50	+38.81	+146.47%	65.70
student_vk	D	88.02	48.10	+39.92	+83.00%	87.86
student_vk	L	40.69	52.70	-12.01	-22.78%	40.00
student_vk	R	85.88	85.70	+0.18	+0.21%	86.06
student_vk	I/VK+D	89.51	45.30	+44.21	+97.59%	89.36
student_vk	I/VK+L	68.48	35.20	+33.28	+94.56%	69.01
student_vk	I/VK+R	92.75	73.40	+19.35	+26.36%	92.82
student_vk	D+L	88.53	51.60	+36.93	+71.57%	88.40
student_vk	D+R	96.45	79.80	+16.65	+20.86%	96.44
student_vk	L+R	89.43	88.70	+0.73	+0.82%	89.48
student_vk	I/VK+D+L	89.68	48.70	+40.98	+84.15%	89.55
student_vk	I/VK+D+R	96.56	70.70	+25.86	+36.57%	96.55
student_vk	I/VK+L+R	92.75	77.80	+14.95	+19.22%	92.83
student_vk	D+L+R	96.44	80.50	+15.94	+19.81%	96.44
student_vk	I/VK+D+L+R	96.53	72.20	+24.33	+33.70%	96.52
student_vk_random	I/VK	65.31	26.50	+38.81	+146.47%	65.70
student_vk_random	D	88.02	48.10	+39.92	+83.00%	87.86
student_vk_random	L	40.80	52.70	-11.90	-22.59%	40.10
student_vk_random	R	85.88	85.70	+0.18	+0.21%	86.06
student_vk_random	I/VK+D	89.51	45.30	+44.21	+97.59%	89.36
student_vk_random	I/VK+L	68.54	35.20	+33.34	+94.73%	69.05
student_vk_random	I/VK+R	92.75	73.40	+19.35	+26.36%	92.82
student_vk_random	D+L	88.52	51.60	+36.92	+71.54%	88.39
student_vk_random	D+R	96.45	79.80	+16.65	+20.86%	96.44
student_vk_random	L+R	89.51	88.70	+0.81	+0.91%	89.57
student_vk_random	I/VK+D+L	89.62	48.70	+40.92	+84.01%	89.49
student_vk_random	I/VK+D+R	96.56	70.70	+25.86	+36.57%	96.55
student_vk_random	I/VK+L+R	92.73	77.80	+14.93	+19.19%	92.81
student_vk_random	D+L+R	96.44	80.50	+15.94	+19.80%	96.43
student_vk_random	I/VK+D+L+R	96.56	72.20	+24.36	+33.74%	96.55
student_vk_cross_scene	I/VK	57.99	26.50	+31.49	+118.82%	57.53
student_vk_cross_scene	D	87.41	48.10	+39.31	+81.73%	87.30
student_vk_cross_scene	L	19.98	52.70	-32.72	-62.09%	17.51
student_vk_cross_scene	R	80.30	85.70	-5.40	-6.30%	80.17
student_vk_cross_scene	I/VK+D	88.84	45.30	+43.54	+96.11%	88.79
student_vk_cross_scene	I/VK+L	59.39	35.20	+24.19	+68.72%	58.81
student_vk_cross_scene	I/VK+R	88.76	73.40	+15.36	+20.93%	88.65
student_vk_cross_scene	D+L	87.65	51.60	+36.05	+69.86%	87.49
student_vk_cross_scene	D+R	94.94	79.80	+15.14	+18.97%	94.93
student_vk_cross_scene	L+R	81.15	88.70	-7.55	-8.51%	81.42
student_vk_cross_scene	I/VK+D+L	88.95	48.70	+40.25	+82.64%	88.88
student_vk_cross_scene	I/VK+D+R	95.32	70.70	+24.62	+34.82%	95.32
student_vk_cross_scene	I/VK+L+R	88.33	77.80	+10.53	+13.53%	88.34
student_vk_cross_scene	D+L+R	94.87	80.50	+14.37	+17.86%	94.87
student_vk_cross_scene	I/VK+D+L+R	95.25	72.20	+23.05	+31.92%	95.25
student_vk_cross_subject	I/VK	65.22	26.50	+38.72	+146.13%	67.65
student_vk_cross_subject	D	90.05	48.10	+41.95	+87.21%	89.96
student_vk_cross_subject	L	38.46	52.70	-14.24	-27.03%	36.69
student_vk_cross_subject	R	85.21	85.70	-0.49	-0.58%	85.45
student_vk_cross_subject	I/VK+D	91.58	45.30	+46.28	+102.17%	91.51
student_vk_cross_subject	I/VK+L	68.95	35.20	+33.75	+95.88%	70.90
student_vk_cross_subject	I/VK+R	93.12	73.40	+19.72	+26.87%	93.12
student_vk_cross_subject	D+L	90.28	51.60	+38.68	+74.97%	90.17
student_vk_cross_subject	D+R	95.71	79.80	+15.91	+19.93%	95.68
student_vk_cross_subject	L+R	88.10	88.70	-0.60	-0.67%	88.14
student_vk_cross_subject	I/VK+D+L	91.72	48.70	+43.02	+88.33%	91.64
student_vk_cross_subject	I/VK+D+R	96.10	70.70	+25.40	+35.92%	96.07
student_vk_cross_subject	I/VK+L+R	93.06	77.80	+15.26	+19.62%	93.06
student_vk_cross_subject	D+L+R	95.74	80.50	+15.24	+18.94%	95.71
student_vk_cross_subject	I/VK+D+L+R	96.12	72.20	+23.92	+33.13%	96.10
student_nv	D	88.57	48.10	+40.47	+84.14%	88.47
student_nv	L	26.31	52.70	-26.39	-50.07%	25.14
student_nv	R	83.37	85.70	-2.33	-2.72%	83.57
student_nv	D+L	88.66	51.60	+37.06	+71.82%	88.55
student_nv	D+R	96.18	79.80	+16.38	+20.53%	96.19
student_nv	L+R	85.39	88.70	-3.31	-3.73%	85.47
student_nv	D+L+R	96.11	80.50	+15.61	+19.39%	96.12
student_nv_random	D	88.57	48.10	+40.47	+84.14%	88.47
student_nv_random	L	26.12	52.70	-26.58	-50.43%	25.01
student_nv_random	R	83.37	85.70	-2.33	-2.72%	83.57
student_nv_random	D+L	88.68	51.60	+37.08	+71.85%	88.56
student_nv_random	D+R	96.18	79.80	+16.38	+20.53%	96.19
student_nv_random	L+R	85.30	88.70	-3.40	-3.83%	85.38
student_nv_random	D+L+R	96.12	80.50	+15.62	+19.41%	96.13
student_nv_cross_scene	D	88.82	48.10	+40.72	+84.65%	88.79
student_nv_cross_scene	L	8.94	52.70	-43.76	-83.03%	6.39
student_nv_cross_scene	R	79.70	85.70	-6.00	-7.01%	79.99
student_nv_cross_scene	D+L	88.92	51.60	+37.32	+72.32%	88.87
student_nv_cross_scene	D+R	95.45	79.80	+15.65	+19.62%	95.45
student_nv_cross_scene	L+R	79.68	88.70	-9.02	-10.16%	79.99
student_nv_cross_scene	D+L+R	95.41	80.50	+14.91	+18.52%	95.40
student_nv_cross_subject	D	90.30	48.10	+42.20	+87.73%	90.21
student_nv_cross_subject	L	25.30	52.70	-27.40	-51.99%	22.23
student_nv_cross_subject	R	84.57	85.70	-1.13	-1.31%	84.75
student_nv_cross_subject	D+L	90.43	51.60	+38.83	+75.26%	90.34
student_nv_cross_subject	D+R	95.76	79.80	+15.96	+20.00%	95.74
student_nv_cross_subject	L+R	86.74	88.70	-1.96	-2.21%	86.76
student_nv_cross_subject	D+L+R	95.70	80.50	+15.20	+18.88%	95.67

来源	组合	Ours Acc	X-Fi Acc	Δ	Rel.Imp	Macro-F1
ablation_no_distill	I/VK	60.94	26.50	+34.44	+129.97%	59.31
ablation_no_distill	D	86.85	48.10	+38.75	+80.55%	86.78
ablation_no_distill	L	33.14	52.70	-19.56	-37.11%	31.11
ablation_no_distill	R	83.13	85.70	-2.57	-3.00%	83.31
ablation_no_distill	I/VK+D	88.09	45.30	+42.79	+94.45%	87.94
ablation_no_distill	I/VK+L	63.63	35.20	+28.43	+80.75%	62.65
ablation_no_distill	I/VK+R	91.05	73.40	+17.65	+24.05%	91.12
ablation_no_distill	D+L	87.24	51.60	+35.64	+69.07%	87.25
ablation_no_distill	D+R	95.57	79.80	+15.77	+19.76%	95.56
ablation_no_distill	L+R	85.92	88.70	-2.78	-3.13%	86.09
ablation_no_distill	I/VK+D+L	88.62	48.70	+39.92	+81.98%	88.53
ablation_no_distill	I/VK+D+R	95.96	70.70	+25.26	+35.73%	95.95
ablation_no_distill	I/VK+L+R	91.23	77.80	+13.43	+17.27%	91.29
ablation_no_distill	D+L+R	95.50	80.50	+15.00	+18.64%	95.50
ablation_no_distill	I/VK+D+L+R	95.99	72.20	+23.79	+32.96%	95.98
ablation_uniform_fusion	I/VK	5.52	26.50	-20.98	-79.16%	1.32
ablation_uniform_fusion	D	85.59	48.10	+37.49	+77.93%	85.30
ablation_uniform_fusion	L	2.65	52.70	-50.05	-94.97%	1.55
ablation_uniform_fusion	R	69.91	85.70	-15.79	-18.42%	68.71
ablation_uniform_fusion	I/VK+D	86.48	45.30	+41.18	+90.90%	86.24
ablation_uniform_fusion	I/VK+L	4.58	35.20	-30.62	-86.98%	1.76
ablation_uniform_fusion	I/VK+R	67.75	73.40	-5.65	-7.69%	66.21
ablation_uniform_fusion	D+L	86.16	51.60	+34.56	+66.98%	85.91
ablation_uniform_fusion	D+R	95.54	79.80	+15.74	+19.73%	95.54
ablation_uniform_fusion	L+R	67.80	88.70	-20.90	-23.56%	66.56
ablation_uniform_fusion	I/VK+D+L	86.43	48.70	+37.73	+77.48%	86.19
ablation_uniform_fusion	I/VK+D+R	95.50	70.70	+24.80	+35.07%	95.49
ablation_uniform_fusion	I/VK+L+R	67.09	77.80	-10.71	-13.76%	65.08
ablation_uniform_fusion	D+L+R	95.42	80.50	+14.92	+18.54%	95.42
ablation_uniform_fusion	I/VK+D+L+R	95.39	72.20	+23.19	+32.12%	95.39

I 当前缺失项审计

任务	类型	项目	状态
HPE	model	ablation/no_kd/best.pth	MISSING_MODEL
HPE	model	ablation/output_kd/best.pth	MISSING_MODEL
HPE	model	ablation/output_token_kd/best.pth	MISSING_MODEL
HPE	model	ablation/output_bone_kd/best.pth	MISSING_MODEL
HPE	model	ablation/full_structural_kd/best.pth	MISSING_MODEL
HPE	model	ablation/uniform/best.pth	MISSING_MODEL
HPE	model	ablation/attention/best.pth	MISSING_MODEL
HPE	model	ablation/uncertainty/best.pth	MISSING_MODEL
HPE	model	ablation/mlp_vk/best.pth	MISSING_MODEL
HPE	model	ablation/skeleton_prompt/best.pth	MISSING_MODEL
HPE	model	teacher_full_cross_scene_split/best.pth	MISSING_MODEL
HPE	model	student_vk_missing_cross_scene_split/best.pth	MISSING_MODEL
HPE	model	student_nv_missing_cross_scene/best.pth	MISSING_MODEL
HPE	model	baseline_full_cross_scene_split/best.pth	MISSING_MODEL
HPE	model	teacher_full_cross_subject_split/best.pth	MISSING_MODEL
HPE	model	student_vk_missing_cross_subject_split/best.pth	MISSING_MODEL
HPE	model	student_nv_missing_cross_subject/best.pth	MISSING_MODEL
HPE	model	baseline_full_cross_subject_split/best.pth	MISSING_MODEL
HPE	eval	student_nv_nonvisual_combinations.csv	MISSING_EVAL
HPE	eval	baseline_full_all_combinations.csv	MISSING_EVAL
HPE	eval	ablation_distillation_no_kd_all_combinations.csv	MISSING_EVAL
HPE	eval	ablation_distillation_output_kd_all_combinations.csv	MISSING_EVAL
HPE	eval	ablation_distillation_output_token_kd_all_combinations.csv	MISSING_EVAL
HPE	eval	ablation_distillation_output_bone_kd_all_combinations.csv	MISSING_EVAL
HAR	model	student_vk_missing_cross_subject/best.pth	MISSING_MODEL

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